MATH 321 — INTRODUCTION TO NUMBER THEORY

Homework 3: Divisibility, Primes, and Congruences

Name: ______________________________ Date: ______________

Instructions. Show all work clearly. Justify each step. Write your solutions in the spaces provided; use the back of the page or attach extra sheets if needed.

1. Divisibility

Definition – Divisibility

Let $a, b \in \mathbb{Z}$ with $a \neq 0$. We say a *divides* b, written $a \mid b$, if there exists an integer k such that $b = ak$. We also say that a is a **divisor** (or **factor**) of b, and that b is a **multiple** of a.

Theorem – Division Algorithm

For any integers a and b with $b > 0$, there exist unique integers q (the *quotient*) and r (the *remainder*) such that

$$a = bq + r, \qquad 0 \leq r < b.$$

Problem 1. Prove that if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.

Hint

Write $b = ak_1$ and $c = ak_2$ for some integers k_1, k_2. What can you say about $b + c$?

Write your answer here.

Problem 2. Use the Division Algorithm to find the quotient and remainder when $a = 2\,027$ is divided by $b = 37$. Verify your answer.

Write your answer here.

Problem 3. Prove that the square of any integer is of the form $3k$ or $3k+1$ for some integer k.

Hint

Consider the three cases $n \equiv 0, 1, 2 \pmod 3$ and square each.

Write your answer here.

2. Prime Numbers and the Fundamental Theorem of Arithmetic

Definition – Prime and Composite

An integer $p > 1$ is called **prime** if its only positive divisors are 1 and p itself. An integer $n > 1$ that is not prime is called **composite**.

Fundamental Theorem of Arithmetic

Every integer $n > 1$ can be written uniquely (up to the order of the factors) as a product of primes:

$$n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}, \qquad p_1 < p_2 < \cdots < p_k, \quad a_i \geq 1.$$

Problem 4. Find the prime factorization of each of the following:

(a) 3 780

(b) 10! (i.e. 10 factorial)

(c) $2^{12} - 1$

Write your answer here.

Problem 5. Prove that there are infinitely many prime numbers.

Hint (Euclid's argument)

Suppose there are only finitely many primes $p_1, p_2, \ldots, p_n$. Consider the number $N = p_1 p_2 \cdots p_n + 1$. Is N divisible by any p_i?

Write your answer here.

Problem 6. Let p be a prime and let $a, b \in \mathbb{Z}$. Prove that if $p \mid ab$, then $p \mid a$ or $p \mid b$.

Hint

This is sometimes called *Euclid's Lemma.* If $p \nmid a$, what is $\gcd(p, a)$? Use Bézout's identity.

Write your answer here.

3. Greatest Common Divisor and the Euclidean Algorithm

Definition – GCD

The **greatest common divisor** of two integers a and b, not both zero, denoted $\gcd(a, b)$, is the largest positive integer d such that $d \mid a$ and $d \mid b$.

Bézout's Identity

For any integers a and b, not both zero, there exist integers x and y such that

$$\gcd(a, b) = ax + by.$$

Problem 7. Use the Euclidean Algorithm to compute $\gcd(252, 198)$. Then find integers x and y such that $252x + 198y = \gcd(252, 198)$.

Hint

Apply repeated division: $252 = 198 \cdot q_1 + r_1$, then $198 = r_1 \cdot q_2 + r_2$, and so on until the remainder is 0. Back-substitute to find x and y.

Write your answer here.

Problem 8. Prove that if $\gcd(a, b) = 1$ and $\gcd(a, c) = 1$, then $\gcd(a, bc) = 1$.

Hint

By Bézout, write $ax_1 + by_1 = 1$ and $ax_2 + cy_2 = 1$. Multiply these two equations together.

Write your answer here.

4. Congruences

Definition – Congruence

Let m be a positive integer. We say a is **congruent** to b **modulo** m, written

$$a \equiv b \pmod{m},$$

if $m \mid (a - b)$.

Fermat's Little Theorem

If p is a prime and $\gcd(a, p) = 1$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Problem 9. Compute each of the following. Show your reasoning.

(a) $3^{100} \pmod{7}$

(b) $2^{340} \pmod{11}$

(c) The last two digits of 7^{999}.

Hint

For (a) and (b), use Fermat's Little Theorem. For (c), work modulo 100 and find the pattern of powers of 7.

Write your answer here.

Problem 10. Solve the linear congruence $17x \equiv 3 \pmod{29}$.

Hint

Find the inverse of 17 modulo 29 using the Extended Euclidean Algorithm or by noting $17 \cdot (\,?\,) \equiv 1 \pmod{29}$.

Write your answer here.

Bonus Challenge

Problem 11 (Bonus). Prove **Wilson's Theorem**: if p is a prime, then

$$(p-1)! \equiv -1 \pmod{p}.$$

Hint

Pair each element $a \in \{1, 2, \ldots, p-1\}$ with its multiplicative inverse $a^{-1} \pmod{p}$. Which elements are their own inverses?

Write your answer here.

End of Homework 3